

Dynamics of a drop at a liquid/solid interface in simple shear fields: A mesoscopic simulation study

Janette L. Jones, Moti Lal, J. Noel Ruddock and Neil A. Spenley

Unilever Research Port Sunlight, Bebington, Wirral, UK L63 3JW

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The dynamics of a surface-confined drop in a simple shear field has been studied, pursuing the dissipative particle dynamics (DPD) simulation approach. The shear field induces contact angle hysteresis in the drop, the degree of hysteresis increasing with the shear rate. At shear rates exceeding a critical value, the drop acquires the tendency to lift off the boundary, leading to its removal. In the equilibrium contact angle range, $\theta^e > 120^\circ$, the drop preserves its integrity as it escapes from the boundary, whereas at lower contact angles the drop assumes a distinctly elongated shape prior to its removal, which develops “necks” at subsequent times. The drop breaks up as the necks are ruptured upon thinning, with some fragments escaping into the bulk phase and some remaining at the surface. Under certain hydrodynamic conditions the moving drop sheds a trail of tiny droplets on the surface. The simulation results are in qualitative agreement with experimental studies on the corresponding systems published in the literature.

1 Introduction

The motion of drops at liquid/solid interfaces, actuated by external forces, constitutes an important component of the mechanisms that drive key processes in detergency, personal products application, oil recovery and surface coating, among many other technologies. The physics of this motion and its consequences for the shape and stability of the drop is not well understood, due partly to the inadequacy of conventional theoretical and computational approaches for treating multiphase systems involving moving boundaries, and partly to the lack of accurate experimental studies on relevant systems. The result is insufficient progress in the development of *new, fundamental* insights into the role played by the gravitational, centrifugal, hydrodynamic and impact forces in processes like the removal of oily soil from fabrics in washing, spreading of creams and lotions on skin and break-up of emulsion drops in the regions close to mixer blades.

The past few years have witnessed the emergence of computer simulation approaches involving time and length scales that are neither atomistic nor phenomenological but lie in a regime residing between the molecular and colloidal domains: the *mesoscopic* regime.¹ With considerable success achieved in the study of the kinetics of domain growth in aggregation processes,² spinodal decomposition,^{3,4} flow through complex microporous media⁵ and rheology of polymers⁶ and colloids,⁷ such approaches are showing significant promise in the modelling of the dynamics of complex systems under hydrodynamic conditions well beyond the Stokes regime. It is thus opportune to consider the application of mesoscopic simulation in the study of the problem of our present concern.

Mesoscopic simulation methods are mainly of two types, lattice based and particle dynamics based. The former represent essentially the revival of the lattice gas (LG) models introduced in the

early years of computer simulation.⁸ In such models the structural elements of the systems are assumed to be located at the nodes of a lattice, the elements moving from node to node. A move involves two stages: collision and propagation, governed by a set of rules consistent with the requirement of mass and momentum conservation at each step. The LG models have been shown to obey the classical Navier–Stokes hydrodynamics equation.⁹ A more powerful simulation approach, based on the idea of solving the Boltzmann transport equation on a lattice (the lattice–Boltzmann method), was proposed a decade ago by McNamara and Zannetti¹⁰ and also by Higuera and Jimenez.¹¹ Further advances have been made in the lattice–Boltzmann (LB) approach leading to important applications with remarkable success.¹² In the particle dynamics based methods, the models assumed are free from lattice structures, the structural elements moving in free space under a variety of forces acting on them. The dynamics of the system is computed at a large number of successive time steps by solving numerically Newton’s equation of motion for each structural element.

In our study, we have pursued the dissipative particle dynamics (DPD) method—a particle dynamics based approach—for modelling the dynamic behaviour of a drop in a simple shear field, located at a solid/fluid boundary. Our objective in these simulations is to establish the mode of deformation, motion and possible break-up of the drop at flow rates corresponding to Reynolds numbers of the order of 1–10, the hydrodynamic regime conveniently accessible by the DPD method. In this paper we give a brief account of the simulation methodology, describe the present model and present our results, delineating the new effects revealed by the simulations.

2 A brief description of the dissipative particle dynamics (DPD) model

The structural element in mesoscopic models is a *momentum carrier*, a conceptual entity whose motion is assumed to represent the collective dynamic behaviour of a large number of molecules in the system. Thus a mesoscopic model is a highly coarse-grained description of matter with each structural element standing for an undefined but a very large number of molecules.

Originally advanced by Koelman and Hoogerbrugge,¹³ DPD is based on the idea of thermostating a simulation “box” by invoking dissipative and fluctuation forces, counter-balancing each other to ensure constant thermal energy in the system. In addition, conservative, *i.e.*, static, forces can be included. Thus the total force acting on a momentum carrier i at any instant is the sum of three forces, dissipative ($\mathbf{F}_i^D$), fluctuation ($\mathbf{F}_i^R$) and conservative ($\mathbf{F}_i^C$):

$$\mathbf{F}_i = \mathbf{F}_i^D + \mathbf{F}_i^R + \mathbf{F}_i^C \quad (1)$$

2.1 Dissipative forces

The dissipative, *i.e.*, hydrodynamic, force between a pair of particles is assumed to be directly related to their relative velocity, in analogy with colloidal systems.¹⁴ Thus the dissipative force, $\mathbf{F}_{ij}^D$, exerted on i by a momentum carrier j may be expressed as

$$\mathbf{F}_{ij}^D = -\boldsymbol{\xi}_{ij} \cdot (\mathbf{v}_i - \mathbf{v}_j) \quad (2)$$

where $\boldsymbol{\xi}_{ij}$ is the dissipation (or resistance) tensor, a function of $\mathbf{r}_{ij}$, the vector representing the separation between i and j . $\mathbf{v}_i$ and $\mathbf{v}_j$ are, respectively, the velocities of i and j . The negative sign in front of $\boldsymbol{\xi}_{ij}$ signifies that the dissipation force acts in the direction opposite to that of the relative velocity. To render the model computationally efficient, the functional form for $\boldsymbol{\xi}_{ij}$ assumed in DPD simulations is extremely simple:

$$\boldsymbol{\xi}_{ij} = \gamma \omega_{ij}^D \mathbf{e}_{ij} \mathbf{e}_{ij} \quad (3)$$

where γ is a scalar coefficient characteristic of the system. ω_{ij}^D is the weight function, which specifies the dependence of $\boldsymbol{\xi}_{ij}$ on $\mathbf{r}_{ij}$, the distance between i and j . $\mathbf{e}_{ij}$ is a unit vector coincident with $\mathbf{r}_{ij}$. Substitution of eqn. (3) into eqn. (2) gives

$$\mathbf{F}_{ij}^D = -\gamma \omega_{ij}^D [\mathbf{e}_{ij} \cdot (\mathbf{v}_i - \mathbf{v}_j)] \mathbf{e}_{ij} \quad (4)$$

2.2 Fluctuation forces

The fluctuation forces are assumed to be Brownian-like, *i.e.*, they are random, conforming to a

Gaussian distribution with zero mean and the second moment proportional to the kinetic energy, kT . $\mathbf{F}_{ij}^R$, the fluctuation force on i due to j , may therefore, be written as

$$\mathbf{F}_{ij}^R = \sigma \zeta_{ij} \omega_{ij}^R \mathbf{e}_{ij} \quad (5)$$

where ζ_{ij} is a random number drawn from a Gaussian distribution whose first moment is equal to zero and variance equal to unity. ω_{ij} is the weight function corresponding to the fluctuation forces. The coefficient σ defines the variance of the fluctuation force which must scale as $1/\sqrt{\delta t}$, where δt is a small time interval during which the force would remain constant.

2.3 Conservative forces

Since the time average of the dissipative and fluctuation forces is zero, they do not feature in the equilibrium behaviour of the system, which is governed solely by the conservative forces. A simple but realistic functional form of $\mathbf{F}_{ij}^C$, universally assumed in mesoscopic models, is

$$\mathbf{F}_{ij}^C = \alpha \omega_{ij}^C \cdot \mathbf{e}_{ij} \quad (6)$$

where ω_{ij}^C is the weight function and α is a parameter.

In respect of both dynamic and conservative forces, a DPD model is completely characterised in terms of the parameters γ , σ and α and the weight functions ω_{ij}^D , ω_{ij}^R and ω_{ij}^C . The relationship between α and γ is furnished by the fluctuation-dissipation theorem:¹⁵

$$\sigma^2/2\gamma = kT \quad (7)$$

with

$$(\omega_{ij}^R)^2 = \omega_{ij}^D \quad (8)$$

Taking kT as the unit of energy, eqn. (7) gives

$$\sigma^2 = 2\gamma \quad (9)$$

In the interest of preserving simplicity in the model, ω_{ij} values are assumed to have the following functional form

$$\begin{aligned} \omega_{ij}^R = (\omega_{ij}^D)^{1/2} = \omega_{ij}^C = \omega_{ij} &= [1 - r_{ij}/r_c] \quad \text{for } r_{ij} < r_c \\ &= 0 \quad \text{for } r_{ij} > r_c \end{aligned} \quad (10)$$

r_c , serving as the unit of length, is the “cut-off” distance above which all the three forces vanish. Eqn. (10) represents a repulsive, spring-like, truncated conservative force, which means that the momentum carriers are assumed to be soft spheres. A further assumption is that the forces are pairwise additive, permitting the force acting on i in a system of N momentum carriers to be expressed as

$$\begin{aligned} \mathbf{F}_i &= \sum_j (\mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R + \mathbf{F}_{ij}^C) \\ &= \sum_j [-\gamma \omega_{ij}^2 \{\mathbf{e}_{ij} \cdot (\mathbf{v}_i - \mathbf{v}_j)\} + \sigma \zeta_{ij} (\omega_{ij}) + \alpha \omega_{ij}] \mathbf{e}_{ij} \end{aligned} \quad (11)$$

The summation $\sum_j$ runs over all momentum carriers except i . Eqn. (11), when combined with Newton’s second law of motion, gives

$$\Delta \mathbf{v}_i = \Delta t/m \left[\sum_j [-\gamma \omega_{ij}^2 \{\mathbf{e}_{ij} \cdot (\mathbf{v}_i - \mathbf{v}_j)\} + \sigma \zeta_{ij} (\omega_{ij}) + \alpha \omega_{ij}] \mathbf{e}_{ij} \right] \quad (12)$$

where $\Delta \mathbf{v}_i$ is the change in the velocity of the momentum carrier i during a small time interval Δt , and m is the particle mass, which is taken as the unit of mass. Eqn. (12) constitutes the equation of motion for a DPD system. The key task in the simulation is to solve this equation for all momentum carriers successively over a large number of time steps to compute the dynamic as well as the equilibrium properties, such as correlation functions and thermodynamic quantities.

3 The present systems

The present systems are each comprised of three phases: two immiscible fluids (1, 2) and a solid (3). In principle, eighteen parameters would be needed to characterise all the like and unlike interactions in such a system; they are $\gamma_{11}, \gamma_{22}, \gamma_{33}, \gamma_{12}, \gamma_{13}, \gamma_{23}, \sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{13}, \sigma_{23}, \alpha_{11}, \alpha_{22}, \alpha_{33}, \alpha_{12}, \alpha_{13}$ and α_{23} . This number is, however, reduced to twelve by virtue of the fluctuation-dissipation theorem [eqn. (9)]. The equation of state for the DPD model enables the value of α for the like interactions to be readily calculated from the compressibility of the fluid. For a pair of partially miscible liquids, α_{12} can be calculated from the solubilities of liquid 1 in 2 and *vice versa*.

3.1 The liquid phases

The two immiscible liquids in our systems are water (1) and oil (2). Ideally, we would like to model incompressible liquids, but in DPD, fluids are necessarily compressible to some degree. The greater the density of particles and the like interaction parameter, the more incompressible is the liquid. However, because the interparticle forces are then greater, the time-step must be smaller, and the algorithm is less efficient. The like interaction must therefore be chosen as a compromise between these two requirements.

The miscibility of the two fluids is controlled mainly by the unlike interaction parameter α_{12} , the two phases would be completely miscible if $\alpha_{12} \approx (\alpha_{11} + \alpha_{22})/2$ and almost entirely immiscible if α_{12} exceeds the mean value appreciably. We defer the discussion on the determination of the γ and σ parameters to the section on simulation details.

3.2 The solid phase

In the model the solid phase is “created” by freezing the motion of the momentum carriers, so they are locked in a certain configuration, which may be crystalline or random. The absence of dynamics in the solid phase means that in the simulation runs the equation of motion is solved only for the momentum carriers in the fluid phases, taking account of their static and dynamic interactions with the solid. The parameters corresponding to the fluid/solid conservative interactions are α_{13} (water/solid) and α_{23} (oil/solid), which together with α_{12} determine, to a large extent, the structure and thermodynamics of the three interfaces featuring in the system.

4 Simulation details

Occupying the bottom section of the simulation “box”, which was subjected to the usual periodic boundary conditions, the particles constituting the solid phase were arranged either in a random configuration (in our initial studies mainly) or in a face-centred cubic structure maintained in a stationary, rigid state at all times. The oil phase was introduced as a drop of volume equal to 113 units (r_c^3) (equivalent spherical radius of 3 units) located at the solid boundary. Water occupied the remaining space in the box. All three phases were assumed to be of the same density, equal to 6. Depending on the properties to be calculated, three box sizes were used: $40 \times 20 \times 20$, $40 \times 10 \times 10$, and $10 \times 10 \times 10$, containing, respectively, 96 000, 24 000 and 6 000 particles. All simulations started from random configurations of the momentum carriers in the fluid phases.

The conservative forces were chosen as follows, where phases 1,2,3 are water, oil and solid, respectively. We took $\alpha_{11} = \alpha_{22} = 12.5$ (this means that water and oil are assumed to be of identical isothermal compressibility and density). The water/oil interaction α_{12} was taken to be 80 (given α_{11} and α_{22} , this fixes the water/oil interfacial tension at 13.5). The water/solid and oil/solid parameters α_{13} and α_{23} were varied between 20 and 40 to control the corresponding surface tensions and thus the contact angle.

The equation of motion was integrated numerically for each particle in the box using the “leap-frog” approach, often pursued in MD simulations. The basic algorithm was developed by Groot and Warren¹⁶ who modified the original version to ensure that the acceleration will feature in the position update of the momentum carriers at each step. Presented succinctly by the set of following equations, the algorithm is accurate to second order in the time step, permitting much longer steps to be assumed than in the original Verlet algorithm.

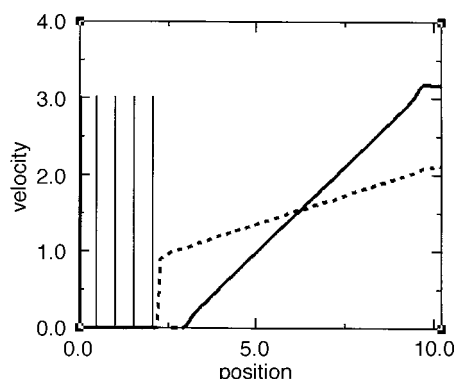


Fig. 1 Computed velocity profiles of the fluid in contact with a solid boundary. The spikes represent the particle layers in the solid substrate. The dashed line is the velocity, showing slip (the velocity does not vanish at the surface). The heavy line is the shear velocity with the modified algorithm. (The units along both axes are arbitrary.)

$$\mathbf{F}_i(t) = f[\mathbf{r}_i(t), \mathbf{v}_i(t - \frac{1}{2} \delta t)]$$

$$\mathbf{v}_i(t) = \mathbf{v}_i(t - \frac{1}{2} \delta t) + \frac{1}{2} \delta t \mathbf{F}_i(t) + O(\delta t)^2$$

$$\mathbf{r}_i(t) = \mathbf{r}_i(t - \delta t) + \delta t \mathbf{v}_i(t) + \frac{1}{2} (\delta t)^2 \mathbf{F}_i(t) + O(\delta t)^3$$

$$\mathbf{v}_i(t + \frac{1}{2} \delta t) = \mathbf{v}_i(t) + \frac{1}{2} \delta t \mathbf{F}_i(t) + O(\delta t)^2 \quad (13)$$

where δt is a small time step during which the force on each momentum carrier would remain virtually constant. Starting from the initial configuration of the momentum carriers, the force on each carrier was calculated. The forces, velocities and positions of the carriers at the next step were computed using the above scheme, which involves the computation of the velocities at half steps (leap-frog). The computation was carried out for a large number of steps, sufficient to cover the duration of the process of interest and to ensure convergence of the time averages of the properties of the system. Positions, velocities and forces of each momentum carrier were recorded at convenient time intervals. From this basic information on the time evolution of the dynamics and configuration of the system, such properties as viscosity, pressure and interfacial tensions were readily obtained.

Groot has carried out an extensive investigation of the effect of the magnitude of δt on the temperature stability of the system.¹⁶ He concluded that the time step of one DPD unit used by Koelman and Hoogerbrugge is much too large to preserve the requisite temperature. The instability problem can only be overcome by reducing the time step to 0.05 DPD units, or to even smaller values if practicable. In the same computations, Groot established through trial and error that the optimum value of σ meeting the requirements of fast temperature equilibration, rapid convergence and stability is around 3.35 for δt in the range from 0.04 to 0.05. In the present simulations we have followed Groot's recommendation in selecting the values for σ . Thus $\sigma_{11} = \sigma_{22} = \sigma_{12} = \sigma_{13} = \sigma_{23} = 3.35$. Hence $\gamma_{11} = \gamma_{22} = \gamma_{12} = \gamma_{13} = \gamma_{23} = 5.6$. The viscosity is determined by the like repulsion α , the dissipation parameter γ and the particle density; for the values chosen, it was equal to 1.74 (for both oil and water).

4.1 Introduction of a shear field

A simple shear flow can be conveniently imposed by subjecting the simulation box to the well known Lees-Edwards boundary condition,¹⁷ whereby the periodic images of the box in the z -direction are moved with velocity equal to $\dot{\gamma}L/2$, where $\dot{\gamma}$ is the shear rate and L the box height. The direction of imposed velocity is positive for the upper image box and negative for the lower image box. The velocities thus imposed propagate into the simulation box producing a linear velocity profile in due course with the plane of zero velocity located in the middle of the box. This boundary condition can be readily adapted to our systems where the surface of the solid boundary

defines the plane of zero velocity. The velocity imparted to the upper image box will, therefore, be equal to $\dot{\gamma}(L - \Delta L)$, where ΔL is the thickness of the solid phase, since the system exists in the $z \geq 0$ semi-infinite space, it includes no lower image boxes.

4.2 Stick boundary condition at the solid/fluid interface

Since the solid phase is maintained in the stationary state, the momentum carriers of the fluid in immediate contact with the solid boundary must, realistically, assume zero velocity ensuring no slip at the fluid/solid interface. However, examination of the velocity profiles obtained in a series of preliminary simulations did show the occurrence of slip; a typical profile is given in Fig. 1. The origin of the slip resided in the repulsive nature of the conservative force depleting the region close to the boundary with the maximum particle density occurring at $z \approx r_c$. Thus most fluid particles remained beyond the range of the fluid/solid dissipative as well as other forces. To eliminate slip, the particles in the region $z < r_c$ were allowed to move freely only in the normal direction. Their velocities in the transverse directions were randomised periodically in such a way that the Maxwell-Boltzmann distribution was preserved, with a mean velocity equal to that of the interface. With this modification the velocity profile (Fig. 1) indicated no slip. This procedure also has the effect of changing the diffusion coefficient of particles near the surface (the surface diffusion coefficient will be somewhat smaller than that in the bulk).

5 Results and discussion

We pursued two separate sets of simulations. In the first set the solid phase was formed by freezing a random configuration of the particles in a slab of 2 unit thickness, and in the second set the solid phase assumed the face-centred cubic structure with the boundary in the 111 plane, the density of both variants of the solid phase being 6. Significant geometrical irregularities in the boundary of the randomly structured solid phase meant that separate simulations had to be carried out over a large number of the frozen configurations to ensure satisfactory convergence in the values of the fluid/solid interface properties. This proved computationally too expensive to be pursued beyond the preliminary stage. Most of the subsequent simulations were, therefore, performed on systems where the solid phase assumed the crystalline structure.

The parameter space investigated here corresponds to $\alpha_{11} = \alpha_{22} = 12.5$ (water and oil are assumed to be of identical isothermal compressibility and density) with a particle density of 6. In this region of the parameter space, the changes in the structure and properties of the fluid/fluid and fluid/solid interfaces are governed entirely by the unlike parameters α_{12} , α_{13} and α_{23} . Properties of our immediate interest are those that essentially determine the configuration of the oil drop at the boundary. These include the various interfacial tensions and the viscosity of the fluid phases.

5.1 Calculation of the interfacial tensions

In simulations the fluid(1)/fluid(2) interfacial tension, Γ_{12} , is conveniently computed using the Irving-Kirkwood equation¹⁸

$$\Gamma_{12} = \int [p_{zz} - \frac{1}{2}(p_{xx} + p_{yy})] dz \quad (14)$$

where p_{xx} , p_{yy} and p_{zz} are the three diagonal components of the pressure tensor. The interface is parallel to the x - y plane. In terms of the interparticle forces eqn. (14) is expressible as

$$\Gamma_{12} = 1/A \left[\sum_{i,j} F_{ijz} r_{ijz} - \frac{1}{2}(F_{ijx} r_{ijx} + F_{ijy} r_{ijy}) \right] \quad i > j \quad (15)$$

$(F_{ijx}, F_{ijy}, F_{ijz})$ and $(r_{ijx}, r_{ijy}, r_{ijz})$ are, respectively, the x , y and z components of the force and the separation vectors between a pair of particles ij , the summation carried over all particle pairs. A is the interface area.

The liquid/solid interface is somewhat more subtle. Assuming, as in the present work, that the interface is modelled by using particles to represent the liquid and these are acted on by an external potential, which represents the effect of the solid, the interfacial tension is given by the expression:

$$\Gamma_{LS} = 1/A \left[\sum_{i,j} F_{ijz} r_{ijz} - \frac{1}{2} (F_{ijx} r_{ijx} + F_{ijy} r_{ijy}) \right] + 1/A \sum_i F_{iz}^{\text{ex}} r_{iz} \quad (16)$$

The second sum is taken over all particles. F_{iz}^{ex} is the force on i due to the external potential. This result was originally derived for a potential constant in the x and y directions¹⁹ but was later shown also to be valid for a potential which is periodic in x and y .²⁰ For systems involving two interfaces, we need to include a third summation in eqn. (16) to account for the second external potential:

$$\Gamma_{LS} = 1/A \left[\sum_{i,j} F_{ijz} r_{ijz} - \frac{1}{2} (F_{ijx} r_{ijx} + F_{ijy} r_{ijy}) \right] + 1/A \sum_i F_{iz}^{\text{ex}1} r_{iz1} + 1/A \sum_i F_{iz}^{\text{ex}2} r_{iz2} \quad (17)$$

where $F_{iz}^{\text{ex}1}$ and $F_{iz}^{\text{ex}2}$ are, respectively, the forces acting on i due to the external potentials corresponding to boundary 1 and boundary 2. r_{iz1} and r_{iz2} are the distances of i from the two boundaries. A is now the combined area of both surfaces. For a hard wall, the position of the interface is defined unambiguously. For any other potential, an arbitrary choice must be made. This corresponds to the usual situation with potential energies, which are always defined relative to some fixed point. Since only differences of free energies are physically meaningful, there is no fundamental ambiguity. In the present model, where the solid boundary is constituted of immobilised particles, the forces on the fluid particles deriving from the external potentials are substituted by those arising from their interaction with the boundary particles. Eqn. (17) thus modifies to

$$\Gamma_{LS} = 1/A \left[\sum_{i,j} F_{ijz} r_{ijz} - \frac{1}{2} (F_{ijx} r_{ijx} + F_{ijy} r_{ijy}) \right] + 1/A \sum_{\alpha} \sum_i F_{i\alpha z} r_{i\alpha z} + 1/A \sum_{\beta} \sum_i F_{i\beta z} r_{i\beta z} \quad (18)$$

Here the variable α runs over all immobile particles in interface A , and β over interface B . In the simulations on the fluid/fluid and the fluid/solid systems forming planar interfaces, ensemble averages of the various summations appearing in eqns. (15) and (18) were computed to obtain the respective interfacial tensions. The results are presented in Fig. 2 where Γ_{12} and Γ_{13} are plotted against α_{12} and α_{13} ($= \alpha_{23}$). For α_{12} , $\alpha_{13} < 10$, the interfacial tensions are negligibly small, rapidly rising to plateau values as the values of these parameters approaches 100. As one would expect, the fluid/solid interfacial tension is always greater than the fluid/fluid interfacial tension for the

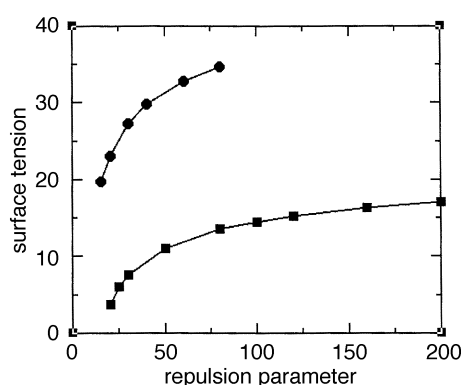


Fig. 2 Plots of computed Γ_{12} , Γ_{13} against α_{12} , α_{13} . ■: Γ_{12} ; ●: Γ_{13} . (The units along both axes are arbitrary.)

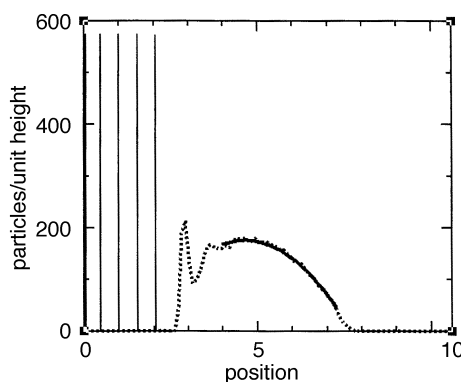
5.2 Shear viscosity, η

$$\begin{aligned} \Pi_{xz} = (1/V) \left\langle \sum_{i,j} r_{ijx} F_{ijz} \right\rangle + (m/V) \left\langle \sum_i (v_{ix} - v_{ix}^s) \cdot (v_{iz} - v_{iz}^s) \right\rangle \\ \downarrow \qquad \qquad \qquad \downarrow \\ \text{virial} \qquad \qquad \qquad \text{kinetic} \end{aligned} \quad (19)$$

5.3 Drop profiles and contact angles in the absence of flow

Fig. 3 shows simulation data and the fit. Note the oscillations caused by the layering of particles near the surface. It is necessary to make an arbitrary choice for the position of the interface; we have taken this to be the first layer of particles in the substrate. Another obvious choice would be the position of the first density maximum in the liquid, but in practice it seems to make little difference.

Table 1 presents the values of contact angles obtained from simulations and predicted by Young's equation from interfacial tensions, for several different sets of parameter values. In general, the agreement is quite good, except for the very lowest contact angle (30°), for which the boundary layer at the substrate probably causes a large distortion.



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Table 1 The drop contact angles measured from the simulation and calculated from Young's equation

α_{13}^a	α_{23}^b	Contact angle (simulation)/degrees	Contact angle (Young)/degrees
80	20	44 (39°)	31
40	20	68	60
40	40	91	90
20	40	114	120
20	80	145	149

^a Repulsion strength for water/solid interaction. ^b Repulsion strength for oil/solid interaction. ^c Computations performed for a drop of an equivalent radius of 9 units.

5.4 Flow-induced deformation and lateral motion of the drop

A drop was placed on a surface and exposed to a shear flow. In each case, the drop was allowed to reach its equilibrium contact angle before shear was applied. We varied the contact angle and shear rate, with the other quantities fixed as above. The capillary and Reynolds numbers are usually defined as $Ca = \eta a \dot{\gamma} / \Gamma$ and $Re = \rho \dot{\gamma} a^2 / \eta$, where a is the equivalent spherical radius of the drop. For the quantities used here, $Ca = 0.39 \dot{\gamma}$ and $Re = 31 \dot{\gamma}$.

We found that at low shear rates, the drop deforms and travels slowly across the surface. Fig. 4 shows the shape of a longitudinal cross-section of the drop (defined as a contour of constant density), indicating how the drop progressively deforms as the shear rate is increased. It may be noted that the advancing and receding dynamic contact angles are no longer equal, *i.e.*, the system displays shear-induced contact angle hysteresis, in agreement with experimental observations. The distortion of the shape caused by the density oscillations at the surface is also visible. It would be interesting to measure the contact angles as a function of shear rate, but the distortion makes this rather difficult. We have attempted to extrapolate the shape of the drop through the oscillating region to the surface, but this proved to be rather inaccurate.

The modification of Young's equation to account for the presence of an external force on the drop seems to have been first made about seventy five years ago by Adam and Jessop.²¹ They recognised that the action of a force parallel to the boundary would render the advancing and receding contact angles, θ_a and θ_r , unequal, the imbalance in the surface forces, occasioned by the changes in the contact angles, being exactly equal to the applied force at equilibrium. The imbalance is positive at the advancing front and negative at the receding end, which means that the external force per unit length, f^{ex} , must be subtracted from and added to Young's equations corresponding to the advancing and receding angles, respectively:

$$\Gamma_{13} - \Gamma_{12} \cdot \cos \theta_a - \Gamma_{23} - f^{\text{ex}} = 0$$

$$\Gamma_{13} - \Gamma_{12} \cdot \cos \theta_r - \Gamma_{23} + f^{\text{ex}} = 0$$

$$\text{or} \quad (\cos \theta_r - \cos \theta_a) = 2f^{\text{ex}} / \Gamma_{12} \quad (20)$$

Thus the degree of hysteresis, $(\cos \theta_r - \cos \theta_a)$, is proportional to the external force per unit length of the drop-boundary contact perimeter. It has been observed in experimental studies that below a certain value of the capillary number the deformed drop remains stationary.²² In such a

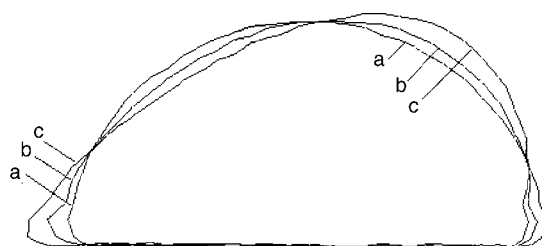


Fig. 4 Cross-section of the drop at Ca equal to 0 (a), 0.04 (b) and 0.08 (c). The equilibrium contact angle is 90° .

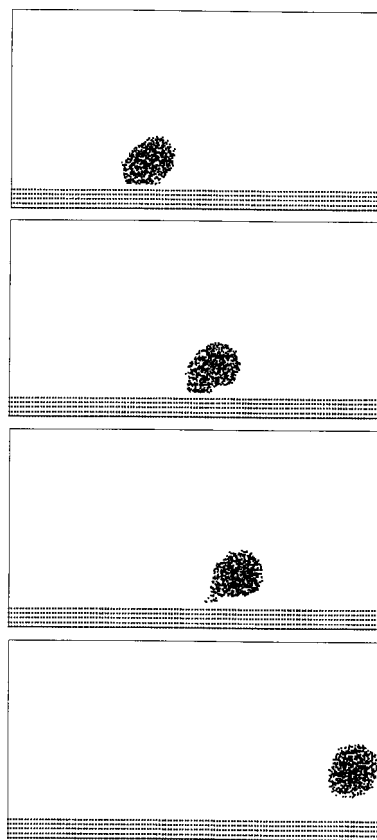


Fig. 5 Removal of a 150° contact angle drop from the surface. The frames are, from top to bottom, at 320, 440, 460 and 500 time units after “switching on” the flow field.

situation, the whole of the external force acting on the drop must be balanced by the extra force on the contact line and, therefore, eqn. (20) is strictly applicable.

In the simulations, the deformed drop moved at all the shear rates applied, resulting in a loss of a fraction of the external force due to viscous dissipation at the drop-boundary interface; so f^{ex} is correspondingly reduced. It follows that the shape of the drop and the extent of the contact angle hysteresis are functions of the drop velocity, which in turn depends on the contact line motion, controlled by a non-classical fluid mechanics traditionally interpreted in terms of a slip length. The friction coefficient between the drop and the surface in the present model is quite large, so at $Ca < 0.1$ (the regime in which the shape profiles in Fig. 4 were computed), the drop moves only slowly. The stress due to the shearing motion is therefore quite small, and the shape should be approximately the same as if the drop were stationary.

Based on their lattice-Boltzmann simulations, Yeomans *et al.*²³ have proposed that for particles moving on solid boundaries diffusion dominates the contact line motion. In our model, the diffusion coefficient near the contact line is determined by the modified boundary condition described above; interestingly, this opens the possibility of changing the surface diffusion coefficient independently of the other parameters in the system.

5.5 Removal of the drop off the boundary

Our simulations show that with increasing shear rates the drop continues to deform and it moves on the surface with increasing velocity. As the shear rate exceeds a certain value, the critical value,

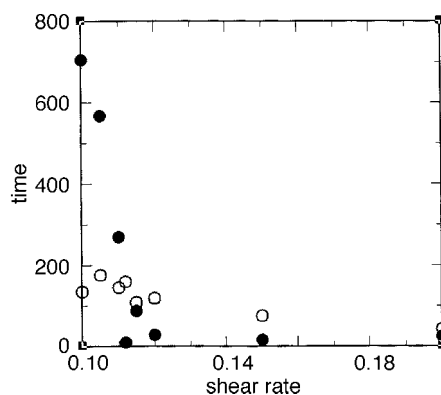


Fig. 6 The fitted parameters τ_0 (open circles) and $\langle \tau_r \rangle$ (filled circles), which define the distribution of removal times, as a function of shear rate. (The units along both axes are arbitrary.)

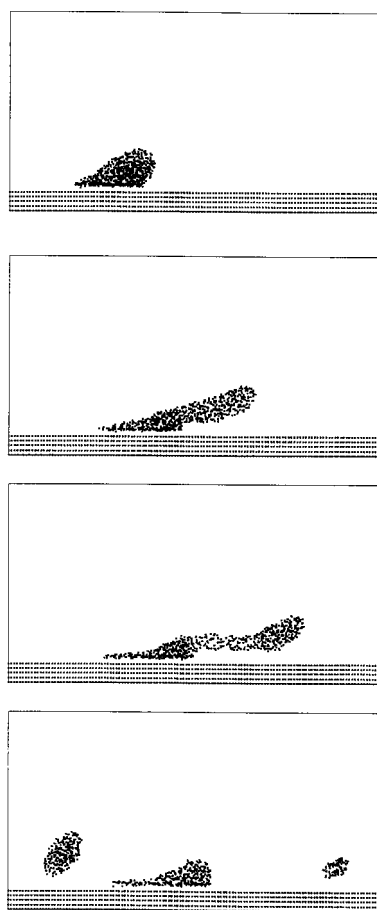


Fig. 7 Break-up and partial removal of a 90° contact angle drop (note that the detached sub-group re-enters from the left through the periodic boundary condition). The snapshots were taken at 155, 200, 210 and 230 time units (from top to bottom).

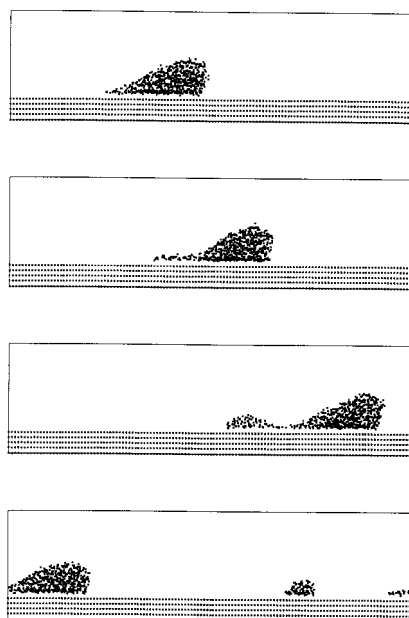


Fig. 8 Trail of droplets left behind a 90° contact angle drop (note periodic boundary condition). The snapshots were taken at 150, 230, 380 and 540 time units (from top to bottom).

the drop develops a tendency to detach itself from the surface, leading to its removal. At shear rates only slightly above the critical value, the deformed drop remains adhered to the surface for a considerable time before being removed. To investigate the removal process, we carried out a large number of simulations at different shear rates. These calculations were performed in a rather small cell ($10 \times 10 \times 10$), since it was necessary to carry out many repeated trials to obtain reliable estimates of the mean time of removal. Fig. 5 presents a series of 4 snapshots showing the lift off process for a drop with equilibrium contact angle equal to 150° .

We found that the removal process possesses a strong stochastic feature whereby the removal times are randomly distributed with a mean value that depends on the shear rate. We have performed a statistical analysis of the data assuming that the mechanism involves two time regimes:

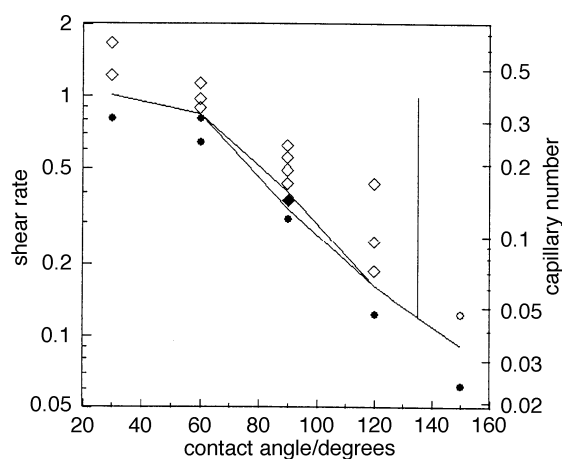


Fig. 9 Observed behaviour of the simulated drop at different contact angles and shear rates. ●: Drop remains on surface; ○: drop removed from surface; ◇: drop partially removed and partially retained; ◆: 'trail of drops' (as in Fig. 8).

one relating to the establishment of the flow in the simulation cell and subsequently inside the drop (τ_0) with attendant deformation of its shape; and the other corresponding to the detachment process (τ_r). It is the second process, which underlies the stochastic nature of the overall removal process. τ_r is assumed to follow an exponential distribution. τ_0 and $\langle \tau_r \rangle$ are plotted against shear rate in Fig. 6. As might be expected, τ_0 varies only weakly with shear rate, while $\langle \tau_r \rangle$ seems to diverge as the critical shear rate is approached.

In explaining the removal process one may invoke the concept of an energy barrier that the drop must overcome in order to escape from the boundary. In the absence of flow the barrier is too large for thermal fluctuations to overcome. The flow field lowers the barrier; at the critical value the barrier height becomes comparable to the thermal energy, so the thermal fluctuations are sufficient to drive the drop away from the surface.

5.6 Drop break-up

In the simulation studies of the drops in the contact angle regime $\leq 120^\circ$, it was observed that they have a strong propensity to deform leading to the development of “necks” in their shape. The necks become unstable upon thinning and are eventually ruptured. This behaviour is similar to that exhibited by drops suspended in strong shear flows.²⁴ The drop may thus break up into several fragments, some remaining on the surface and some escaping into the bulk phase. Fig. 7 depicts the break-up process, where a time sequence of the configuration snapshots from the simulation of a drop at 90° contact angle is presented. Experimental studies on confined drops in shear fields reveal a remarkably similar picture of the break-up process.²⁵ The fraction of the drop removed depends on the contact angle; for example, at 30° contact angle, only a tiny fraction of the drop is removed whereas at 120° most of the drop leaves the surface.

At a contact angle of 90° and intermediate shear rates, we see the drop being dragged along the surface, leaving a trail of small droplets behind it, Fig. 8. This behaviour is of common occurrence and has been studied systematically in several experimental investigations, *e.g.*, ref. 26.

Fig. 9 summarises our simulation results on the state of surface-confined drops in a range of shear rates and contact angles. Each point corresponds to a simulation run. The lines are conjectured boundaries between the different regions.

6 Concluding remarks

In the present simulation study we have successfully modified the DPD algorithm to ensure the occurrence of the stick boundary condition at a solid/fluid interface, a necessary step in the development of a credible mesoscopic model for the three-phase solid/drop/fluid system.

In the absence of flow, the computed values of the equilibrium contact angles, θ^e , are found to be in satisfactory agreement with those given by Young's equation, for $\theta^e > 90^\circ$. The determination of θ^e for values less than 90° proved problematic due to poor definition in the drop profile close to the solid boundary, a result of oscillations in the fluid structure in the interfacial region. In this range of θ^e the large departures observed in the DPD values from those calculated from Young's equation may be possibly ascribed to the lack of requisite accuracy in the former. The problem may be alleviated by increasing the drop size in the simulations, so the magnitude of the profile distortion at the interface is reduced relative to the drop size, leading to better estimates of the contact angles. Alternatively, the quality of the drop profile close to the boundary may be improved by smoothing the density in the interfacial region by modelling the wall as a potential chosen to compensate for the oscillations.²⁷ The disadvantage of this approach is that there is only one such potential and therefore only one possible value of the interfacial tension.

The simulations unequivocally establish the occurrence of shear-induced contact angle hysteresis, in agreement with various experimental results reported in the literature. However, the problems originating from the distortion of the drop profile at the interface precluded accurate computation of the degree of hysteresis, $(\cos \theta_r - \cos \theta_a)$. As stated above, simulation of large drops would, hopefully, permit the desired accuracy to be achieved, enabling validity of the DPD model to be tested through comparison with the results from the Adam–Jessop equation.

The mechanism of the removal of the drop from the boundary involves two time scales corresponding, respectively, to the attainment of steady shear flow and the escape of the drop. The

latter is stochastic in nature. The characteristic time for the stochastic process has been shown to diverge in the limit of the critical shear rate.

The simulation results presented here, although preliminary, are quite encouraging, providing sufficient motivation for extending our modelling studies to systems such as large drops at solid boundaries, drops at rough boundaries and interfaces involving surface tension gradients.

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